Good cutoff functions (Cheeger-Colding)

Let $\mathrm{Ric}_{M^n} \geq -(n-1)$, $0 < R_1 < R_2$. $\exists\, \phi : M^n \to [0,1]$ such that

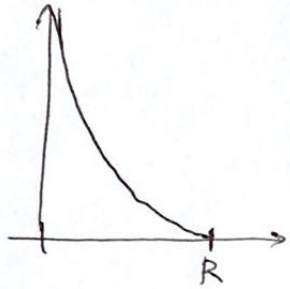
$$\phi = 1 \quad \text{on } B_{R_1}(p)$$
$$\phi = 0 \quad \text{on } M \setminus B_{R_2}(p)$$
$$|\nabla \phi|,\ |\Delta \phi| \leq C(n, R_1, R_2)$$

Proof Recall $\underline{G}, \underline{U}$. Let $\underline{G}_R = \underline{G} - \underline{G}(R)$
$\underline{U}_R = \underline{U} - \underline{U}(R)$

for suitable $c_R > 0$

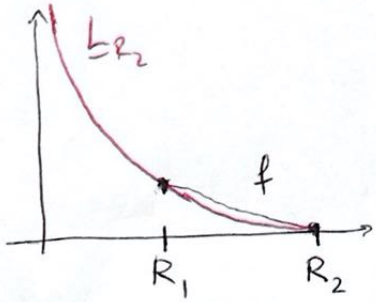
$$\underline{L}_R = c_R \underline{G}_R + \underline{U}_R$$

satisfies
$$\Delta \underline{L}_R = 1$$
$$\underline{L}_R' \leq 0$$
$$\underline{L}_R(R) = 0$$



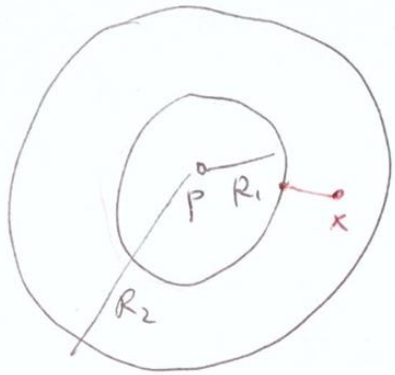
Let $f : \overline{B_{R_2}(p)} \setminus B_{R_1}(p) \to \mathbb{R}$

s.t.
$$\Delta f = 1$$
$$f = \underline{L}_{R_2} \quad \text{on } \partial\left(B_{R_2}(p) \setminus B_{R_1}(p)\right)$$



By Laplacian comparison
$$\Delta \underline{L}_{R_2} \geq 1$$
$$\Rightarrow f \geq \underline{L}_{R_2}$$

By quantitative max principle
$$f(x) \leq \max_{\partial B_{R_1}(p)} \left(f - \underline{U}(\rho_x)\right)$$
$$= \underline{L}_{R_2}(R_1) - \underline{U}(R - R_1), \quad R = d(p,x)$$



$\Rightarrow$ can let $a < b$ s.t.

$$\underline{L}_{R_2}(R_1) - \underline{U}(R_2 - R_1) < a < b < \underline{L}_{R_2}(R_1)$$

and $\eta_1, \eta_2 > 0$ s.t.

$$b = \underline{L}_{R_2}(R_1 + \eta_1)$$

$$a = \underline{L}_{R_2}(R_1) - \underline{U}(R_2 - \eta_2 - R_1)$$

$\Rightarrow$ on $B_{R_1+\eta_1}(p)$ $\quad f \geq b$

on $B_{R_2} \setminus B_{R_2-\eta_2}$ $\quad f \leq a$

Let $\psi(s) = \begin{cases} 1 & \text{for } s \geq b \\ 0 & \text{for } s \leq a \end{cases}$ and $\phi = \psi(f)$

$\phi = 1$ near B_{R_1} and $\phi = 0$ near ∂B_{R_2}

$$\nabla \phi = \psi'(f) \nabla f$$

$$\Delta \phi = \psi''(f) |\nabla f|^2 + \psi'(f) \overset{=1}{\cancel{\Delta f}}$$

Cheng-Yau $\Rightarrow$ $|\nabla f|$ is bounded where $\psi''(f) \neq 0$

$\Rightarrow$ get bounds on $|\nabla \phi|$, $|\Delta \phi|$. $\square$

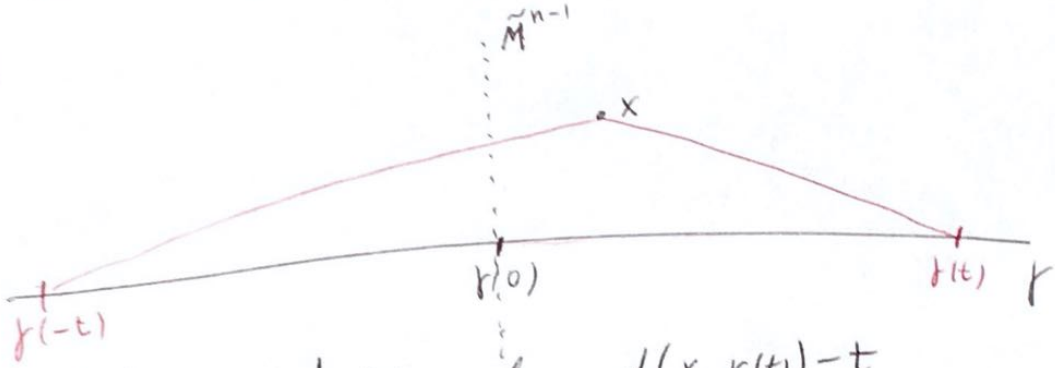
Splitting theorem (Cheeger–Gromoll)

Let $\mathrm{Ric}_{M^n} \geq 0$, M^n complete

$\gamma : \mathbb{R} \to M$ a line $\left(d(\gamma(s), \gamma(t)) = |s-t| \right)$

Then $M^n = \mathbb{R} \times \tilde{M}^{n-1}$ isometrically.

Proof



Define $b_+(x) = \lim_{t \to \infty} d(x, \gamma(t)) - t$

$$\left(\text{by triangle ineq: } \begin{aligned} & d(x, \gamma(s)) - s \\ & \leq d(x, \gamma(t)) - t \\ & \text{if } s > t \end{aligned} \right.$$

$$\left. \text{and } d(x, \gamma(t)) - t \geq -d(x, \gamma(0)) \right)$$

Similarly define $b_-(x) = \lim_{t \to \infty} d(x, \gamma(-t)) - t$

Laplacian comparison

$$\Delta b_\pm \leq 0 \qquad \left(\Delta_x d(x, \gamma(t)) \leq \frac{n-1}{d(x, \gamma(t))} \right)$$

but $b_+(x) + b_-(x) = \lim_{t \to \infty} d(x, \gamma(t)) + d(x, \gamma(-t)) - 2t$

$$\geq 0$$

and $b_+ + b_- = 0$ on γ.

Strong max. principle $\Rightarrow$ $b_+ + b_- \equiv 0$ on M^n,

$$b_+ = -b_- .$$

$$\Delta b_\pm \leq 0 \quad \Rightarrow \quad \Delta b_\pm = 0$$

and $|\nabla b_\pm| = 1$

Bochner $\Rightarrow$ $0 = \frac{1}{2}\Delta |\nabla b_+|^2 = |\mathrm{Hess}_{b_+}|^2 + \underbrace{\mathrm{Ric}(\nabla b_+, \nabla b_+)}_{\geq 0}$

So $\mathrm{Hess}_{b_+} = 0$ i.e. ∇b_+ is parallel.

Can define $\tilde{M}^{n-1} = b_+^{-1}(0)$

and isometry $\tilde{M}^{n-1} \times \mathbb{R} \to M^n$

is given by $(x, t) \mapsto F_t(x)$

F_t = flow along ∇b_+. $\square$

<u>Almost splitting theorem</u> (Cheeger–Colding)

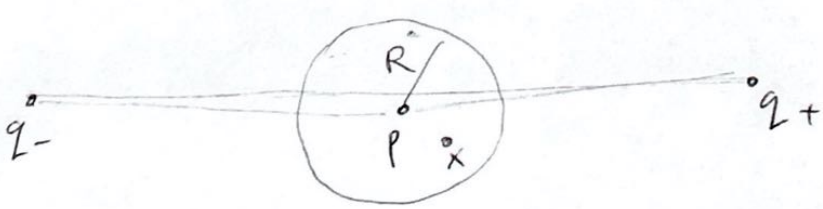
<u>Notation</u>: write $\Psi(\varepsilon_1, \dots, \varepsilon_k \mid c_1, \dots, c_N)$

for nonnegative function s.t.

for fixed $c_1, \dots, c_N$ we have

$\Psi \to 0$ as $\varepsilon_1, \dots, \varepsilon_k \to 0$

<u>Theorem</u> Let $\left.\begin{array}{l} \mathrm{Ric} \geq -(n-1)\delta \\ d(p, q_\pm) \geq L \\ 0 \leq E(p) := d(p, q_+) + d(p, q_-) - d(q_+, q_-) \leq \varepsilon \end{array}\right\} (*)$



Then there is a length space X s.t. for ball $B_R((0,x)) \subset \mathbb{R} \times X$, $x \in X$ we have

$$d_{GH}\left(B_R(p), B_R((0,x))\right) \leq \Psi(\delta, L^{-1}, \varepsilon \mid n, R)$$

<u>Strategy</u> Let $b_\pm(x) = d(x, q_\pm) - d(p, q_\pm)$

<u>Goal</u>: if $\delta, L^{-1}, \varepsilon$ are sufficiently small (given R, n) then $b_\pm$ behave like in the splitting thm.

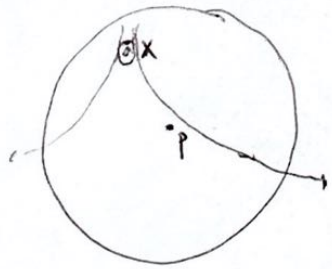
Better to use $h_\pm$ s.t.

$$\Delta h_\pm = 0 \quad \text{on } B_R(p)$$
$$h_\pm = b_\pm \quad \text{on } \partial B_R(p)$$

<u>Theorem</u> (Abresch–Gromoll) Under $(*)$ we have

$$E \leq \Psi = \Psi(\delta, L^{-1}, \varepsilon \mid n, R) \quad \text{on } B_R(p)$$

Ψ will always denote $\Psi(\delta, L^{-1}, \varepsilon \mid n, R)$ but may change from line to line.

<u>Proof</u> Laplacian comparison $\Rightarrow \Delta E \leq \Psi$ on $B_{3R}(p)$

$$E(p) \leq \varepsilon$$
$$|\nabla E| \leq 2$$